

Describing Use Cases [[Browse Marketplace](#)]

ID Number – Name:	UC-1 – Browse Marketplace
Short-Description / Goal:	A user (Guest, Registered User, or Upcycler) browses all available item listings on the marketplace and applies filters, sorting, and search to find relevant items.
Preconditions:	System is online and item listings are available.
Postconditions:	User views a filtered/sorted list of items matching their criteria.
Error Situations – System State in the Event of an Error:	No items match the applied filters – System displays an empty results message with suggestions to broaden the search.
Actor(s) / Initiator(s):	Guest, Buyer, Seller, Swapper, Upcycler
Trigger:	User opens the marketplace or enters a search query.
Standard Process (Main Success Scenario):	(1) User navigates to the Marketplace page. (2) System displays all active listings by default. (3) User applies filters (e.g., category, material, condition, price range, upcycled-only). (4) User applies sorting (e.g., newest, price low-to-high, trust score). (5) System updates and displays the filtered and sorted results. (6) User clicks on an item to view its full details.
Alternative Processes (Alternative/Other Scenarios):	(3') User types a keyword in the search bar – System returns matching results. (5') No results found – System suggests broadening filters or browsing trending items.

Describing Use Cases [[Register](#)]

ID Number – Name:	UC-2 – Register
Short-Description / Goal:	A new visitor creates an account on the platform by providing their information and completing a preference selection to personalize their experience.
Preconditions:	User does not have an existing account. User has a valid email address and password.
Postconditions:	A new account is created and stored. User's preferences are saved and a personalized feed is generated. User is redirected to the home feed.
Error Situations – System State in the Event of an Error:	Email is already registered – System prompts user to log in or reset password. Invalid data format – System highlights the invalid fields without clearing the form.
Actor(s) / Initiator(s):	Guest
Trigger:	Guest clicks the 'Register' button.
Standard Process (Main Success Scenario):	(1) Guest clicks 'Register'. (2) System displays the registration form (name, email, password, location). (3) Guest fills in the required information. (4) System validates all fields and checks email uniqueness. (5) System stores the new account data. (6) System launches the Preference Selection wizard (style aesthetics, sizes, favourite materials, upcycle interest level) — <i>similar to Pinterest onboarding</i>. (7) User completes preference selection. (8) System curates a personalised home feed based on selected preferences. (9) User is redirected to their personalised home feed.
Alternative Processes (Alternative/Other Scenarios):	(4') Email already exists – System notifies user and offers a 'Log In' link. (7') User skips preference selection – System shows a default trending feed and prompts preferences later.

Describing Use Cases [[Login](#)]

ID Number – Name:	UC-3 – Login
Short-Description / Goal:	A registered user authenticates into the platform using their email and password to access their account and perform actions.
Preconditions:	User has a registered account. System is online.
Postconditions:	User is authenticated and redirected to their personalised home feed. Session is created.
Error Situations – System State in the Event of an Error:	Invalid credentials – System displays 'Login Failed' message; account is not locked after first attempt.
Actor(s) / Initiator(s):	Registered User (Buyer, Seller, or Upcycler — role is determined by actions taken, not fixed to the account)
Trigger:	User clicks the 'Login' button on the landing page or is redirected after attempting a restricted action.
Standard Process (Main Success Scenario):	(1) User clicks 'Login'. (2) System displays the login form (email, password). (3) User enters their credentials. (4) System validates credentials against the database. (5) Credentials are correct – System creates a session. (6) System redirects user to their personalised home feed.
Alternative Processes (Alternative/Other Scenarios):	(1') User has no account – System offers a 'Register' link. (4') Credentials are invalid – System displays 'Login Failed' and allows retry.

Describing Use Cases [[Ask for Upcycling Role](#)]

ID Number – Name:	UC-4 – Ask for Upcycling Role
Short-Description / Goal:	A registered user applies for the Upcycler role by submitting a portfolio for admin review, allowing them to list upcycled items and access creator-specific features.
Preconditions:	User is logged in as a standard Registered User. User has a portfolio of upcycling work ready to upload.
Postconditions:	[Approved] User's role is updated to Upcycler; user is notified and gains access to Upcycler features. [Rejected] User is notified with a rejection reason and remains a standard Registered User.
Error Situations – System State in the Event of an Error:	Portfolio upload fails – System displays an upload error and allows retry without losing other form data.
Actor(s) / Initiator(s):	Registered User, System Admin
Trigger:	Registered User clicks 'Apply for Upcycler Role'.
Standard Process (Main Success Scenario):	(1) User navigates to their profile and clicks 'Apply for Upcycler Role'. (2) System displays the application form. (3) User uploads portfolio images/links and submits the application. (4) System notifies the Admin of a pending application. (5) Admin reviews the portfolio. (6) Admin approves the application. (7) System updates the user's role to Upcycler. (8) System sends an approval notification to the user. (9) User can now access Upcycler-specific features (Track Transformation, Upcycle Item, etc.).
Alternative Processes (Alternative/Other Scenarios):	(6') Admin rejects the application – Admin enters a rejection reason. (7') System sends a rejection notification with the reason to the user. (8') User remains a standard Registered User and may reapply.

Describing Use Cases [[Post in Community](#)]

ID Number – Name:	UC-5 – Post in Community
Short-Description / Goal:	A registered user creates a post in the community feed. Post types include 'Style Request' (styling advice on an item) and 'Inspire Me' (finding a similar item). All posts pass through automated fraud detection before going live.
Preconditions:	User is logged in. Community feed is accessible.
Postconditions:	[Accepted] Post is published in the community feed and visible to other users. [Rejected] Post is blocked and user is notified of the violation.
Error Situations – System State in the Event of an Error:	Post content triggers fraud detection – System auto-rejects the post and notifies the user with a reason.
Actor(s) / Initiator(s):	Buyer, Seller, Swapper, Upcycler, Moderator
Trigger:	User clicks 'New Post' in the community feed.
Standard Process (Main Success Scenario):	(1) User clicks 'New Post'.as 'Style Request' or 'Inspire Me'. (1.1) [Style Request] User uploads a photo of an item and asks how to style it. (1.2) [Inspire Me] User uploads a photo and asks where to find a similar item; system checks marketplace first, then post to the community. (2) User submits the post. (3) System runs automated fraud detection on the content (prohibited words, misleading claims). (4) Post passes detection – System publishes the post. (5) Other users can comment with suggestions or recommendations. (6) Post author selects a preferred response; system may reward the idea's owner with Eco-Credits.
Alternative Processes (Alternative/Other Scenarios):	(3') Fraud detection flags the post – System rejects it and notifies the user.

Describing Use Cases [[Manage Community](#)]

ID Number – Name:	UC-6 – Manage Community
Short-Description / Goal:	A Community Moderator oversees the community feed by approving or rejecting posts, handling harassment reports, and resolving community conflicts. Major decisions require System Admin approval.
Preconditions:	Moderator is logged in with Moderator-level permissions. Community content or reports are pending review.
Postconditions:	Flagged content is actioned (approved/rejected/removed). Community conflict is resolved or escalated to Admin.
Error Situations – System State in the Event of an Error:	Moderator attempts an action that requires Admin authorization – System blocks the action and sends an escalation request to Admin.
Actor(s) / Initiator(s):	Community Moderator, System Admin
Trigger:	A post is flagged by a user, fraud detection, or appears in the moderation queue.
Standard Process (Main Success Scenario):	(1) Moderator opens the moderation dashboard. (2) System displays a queue of flagged posts/comments/accounts. (3) Moderator reviews the flagged content. (4) Moderator decides to approve or reject the content. (5) For significant actions (e.g., banning an account), Moderator submits a decision for Admin approval. (6) Admin reviews and approves or overrides the Moderator's decision. (7) System applies the final action and notifies the relevant users.
Alternative Processes (Alternative/Other Scenarios):	(4') Harassment or bullying is confirmed – Moderator escalates directly to Admin for account suspension. (6') Admin overrides Moderator's decision – System logs the override and applies Admin's ruling.

Describing Use Cases [[Mentor a Mentee](#)]

ID Number – Name:	UC-7 – Mentor a Mentee
Short-Description / Goal:	An experienced Upcycler (Mentor) accepts a mentorship request from a novice Upcycler (Mentee), supervises their upcycling project, and earns a badge or Eco-Credit reward upon the Mentee's successful sale.
Preconditions:	Mentor is a verified Upcycler. Mentee is a verified Upcycler working on an upcycling project. Mentee has submitted a mentorship request.
Postconditions:	Mentee receives guidance on their upcycling project. [Upon successful sale] Mentor receives Eco-Credits. System updates both Mentor and Mentee Badge rankings.
Error Situations – System State in the Event of an Error:	Mentor becomes unresponsive – System reassigns the Mentee to another available Mentor after a defined timeout.
Actor(s) / Initiator(s):	Upcycler (Mentor), Upcycler (Mentee)
Trigger:	Mentee clicks 'Request a Mentor' during an upcycling project.
Standard Process (Main Success Scenario):	(1) Mentee initiates an upcycling project and selects 'I need a mentor'. (2) System matches the Mentee with available, high-ranked Mentors. (3) Mentor receives a mentorship request and accepts. (4) Mentor supervises the Mentee's transformation process (reviews before/after photos, gives advice). (5) Mentee completes the project and lists the item. (6) Item is sold successfully. (7) System rewards the Mentor with Eco-Credits. (8) System updates both users' Badge rankings.
Alternative Processes (Alternative/Other Scenarios):	(3') Mentor declines the request – System offers the request to the next available Mentor. (1') Mentee selects 'I did it on my own' – Mentorship flow is skipped; Mentee documents independently.

Describing Use Cases [[Report](#)]

ID Number – Name:	UC-8 – Report
Short-Description / Goal:	A user reports a post, comment, or account for a violation. Financial disputes are escalated to Admin; community-related reports are handled by the Moderator.
Preconditions:	User is logged in. The reported content or account exists in the system.
Postconditions:	Report is logged and routed to the appropriate handler (Admin or Moderator). Appropriate action is taken based on the report type.
Error Situations – System State in the Event of an Error:	Report submission fails due to a system error – System displays a retry option and logs the failure.
Actor(s) / Initiator(s):	Registered User, Upcycler, Community Moderator, System Admin
Trigger:	User clicks 'Report' on a post, comment, or user profile.
Standard Process (Main Success Scenario):	(1) User clicks 'Report' on the target content or account. (2) System displays a report form with reason categories (e.g., Fraud, Harassment, Counterfeit, Misleading Listing). (3) User selects a reason and optionally adds a description. (4) User submits the report. (5) System logs the report and routes it: financial disputes → Admin; community issues → Moderator. (6) Handler reviews and takes action (warn, remove content, suspend account). (7) System notifies the reporting user of the outcome.
Alternative Processes (Alternative/Other Scenarios):	(6') Content is found non-violating – Report is dismissed; reporting user is notified.

Describing Use Cases [[Assess Item](#)]

ID Number – Name:	UC-9 – Assess Item
Short-Description / Goal:	A registered user (Seller or Swapper_A or Upcycler) creates a detailed item listing by providing photos, condition tags, material type, and description. Supports single-item and bulk listing.
Preconditions:	User is logged in as a Registered User or Upcycler. User has an item to list (New or from Virtual Closet).
Postconditions:	Item is listed on the marketplace and visible to other users. Item is available for purchase, swapping, or offers.
Error Situations – System State in the Event of an Error:	Fraud detection flags a misleading description – Listing is rejected; user is notified with a reason. Required fields are missing – System highlights incomplete fields and prevents submission.
Actor(s) / Initiator(s):	Seller, Swapper_A, Upcycler
Trigger:	User clicks 'List an Item' or 'Quick List' from their Virtual Closet.
Standard Process (Main Success Scenario):	(1) User clicks 'List an Item'. (2) System presents a guided listing form. (3) User uploads photos (min. 1 required). (4) User fills in item details: title, category, material type, condition tags (e.g., 'Pilling on sleeves'), size, price. (5) User sets availability: For Sale, For Swap, or Both. (6) User submits the listing. (7) System runs fraud/content moderation check. (8) Listing passes check – System publishes the item to the marketplace.
Alternative Processes (Alternative/Other Scenarios):	(1') User selects 'Bulk Listing' – System allows uploading multiple items at the same time. (7') Fraud check fails – System rejects listing and notifies user.

Describing Use Cases [[Manage Virtual Closet](#)]

ID Number – Name:	UC-10 – Manage Virtual Closet
Short-Description / Goal:	A registered user manages a private digital inventory of clothing they own but have not listed for sale or swap. Items in the Virtual Closet can be used for AI outfit generation or quickly converted to a marketplace listing.
Preconditions:	User is logged in. User has items they wish to add to their private closet.
Postconditions:	Item is saved in the Virtual Closet and available for AI outfit suggestions. [If listed] Item transitions from Virtual Closet to active marketplace listing.
Error Situations – System State in the Event of an Error:	Image upload fails – System prompts user to retry without losing other entered data.
Actor(s) / Initiator(s):	Buyer, Seller, Swapper, Upcycler
Trigger:	User navigates to 'My Closet' from their profile.
Standard Process (Main Success Scenario):	(1) User opens 'My Closet'. (2) System displays all items currently in the Virtual Closet. (3) User clicks 'Add Item' and uploads a photo with basic details (name, material, colour). (4) System saves the item to the closet. (5) Closet items are available as input for the Generate AI Outfit feature.
Alternative Processes (Alternative/Other Scenarios):	(3') User selects an existing closet item and clicks 'List for Sale/Swap' – System pre-fills the Assess Item form with the closet item's details for quick listing. (3'') User removes an item from the closet – System deletes it from the private inventory.

Describing Use Cases [[Generate AI Outfit](#)]

ID Number – Name:	UC-11 – Generate AI Outfit
Short-Description / Goal:	A registered user submits an item (from their Virtual Closet or the marketplace) to an AI assistant that suggests a complete outfit and provides an AR fitting room to visualize the result, reducing returns.
Preconditions:	User is logged in. User has at least one item in their Virtual Closet or has selected a marketplace item.
Postconditions:	AI-generated outfit suggestion is displayed. User can view himself (optionally) in the AR fitting room. User can add suggested items directly to cart.
Error Situations – System State in the Event of an Error:	AI service is unavailable – System displays an error message and offers to post a Style Request to the community instead.
Actor(s) / Initiator(s):	Registered User, Upcycler
Trigger:	User clicks 'Generate Outfit' on an item in their closet or on a marketplace listing.
Standard Process (Main Success Scenario):	(1) User selects an item and clicks 'Generate Outfit'. (2) System sends the item and user's preference profile to the AI engine. (3) AI suggests a complete outfit by combining closet items and/or marketplace listings. (4) User views the outfit suggestion. (5) User tries the outfit in the AR Fitting Room. (6) User adds preferred items to cart or saves the outfit.
Alternative Processes (Alternative/Other Scenarios):	(3') AI cannot find enough matching items – System suggests posting a 'Style Request' to the community. (5') User skips AR and proceeds directly to adding items to cart.

Describing Use Cases [[Track Transformation](#)]

ID Number – Name:	UC-12 – Track Transformation
Short-Description / Goal:	An Upcycler documents the transformation of a garment by uploading before/after photos, describing modifications made, logging additional materials used, and sharing an Upcycling Story to build credibility and trust.
Preconditions:	User is logged in as an Upcycler. User has an item undergoing or completed an upcycling transformation.
Postconditions:	Transformation log is saved and attached to the item listing. Carbon savings and Eco-Credits are calculated and displayed. Upcycling Story is visible on the item listing page.
Error Situations – System State in the Event of an Error:	Photo upload fails – System allows partial save and prompts user to complete the log before publishing.
Actor(s) / Initiator(s):	Upcycler
Trigger:	Upcycler starts a new upcycling project or opens an existing draft transformation log.
Standard Process (Main Success Scenario):	(1) Upcycler creates a new transformation log. (2) Upcycler uploads 'Before' photo(s) of the original garment. (3) Upcycler describes the modifications made (e.g., 'Removed sleeves, added embroidery, re-dyed with natural dye'). (4) Upcycler logs additional materials used (type, quantity). (5) Upcycler uploads 'After' photo(s) of the finished item. (6) Upcycler writes the Upcycling Story. (7) System calculates carbon savings based on material type and weight. (8) System awards Eco-Credits based on impact score. (9) Transformation log is saved and linked to the item listing.
Alternative Processes (Alternative/Other Scenarios):	(4') Upcycler indicates zero additional materials used – System records it as a repair-only transformation. (6') Upcycler skips the story – System saves the log without a story; story can be added later.

Describing Use Cases [[Upcycle Item](#)]

ID Number – Name:	UC-13 – Upcycle Item
Short-Description / Goal:	An Upcycler selects an upcycling path: (A) transform an owned item to resell for profit (Flip for Profit), or (B) perform an upcycling service on a client's item and return it. Path B requires a Security Bond to protect the client.
Preconditions:	User is logged in as a verified Upcycler. Item to be upcycled is available (owned by Upcycler or submitted by a client).
Postconditions:	[Path A] Upcycled item is listed for sale on the marketplace with a transformation log. [Path B] Upcycled item is returned to the client; Security Bond is released.
Error Situations – System State in the Event of an Error:	[Path B] Item is damaged during the service – Security Bond is used to compensate the client; trust score is reduced.
Actor(s) / Initiator(s):	Upcycler, Registered User (as client in Path B)
Trigger:	Upcycler clicks 'Start Upcycling Project'.
Standard Process (Main Success Scenario):	(1) Upcycler clicks 'Start Upcycling Project'. (2) System asks: 'Do you already own the piece?' or 'Are you providing a service?' (3a) [Path A – Flip for Profit] Upcycler selects their item (owned or to be purchased from the site). (4a) Upcycler performs the upcycling work and logs the transformation (see UC-12). (5a) Upcycler lists the finished item for sale on the marketplace. (3b) [Path B – Service] Client submits the item and pays a Security Bond (equivalent to item's estimated value). (4b) Upcycler performs the service and logs the transformation. (5b) Upcycler returns the item to the client. (6b) Client confirms receipt – Security Bond is released to the Upcycler.
Alternative Processes (Alternative/Other Scenarios):	(3a') Upcycler buys an item from the marketplace to upcycle – Standard Buy Item flow is triggered first. (5b') Client disputes the work quality – Admin reviews and decides on Bond release or partial refund.

Describing Use Cases [[Sell Item](#)]

ID Number – Name:	UC-14 – Sell Item
Short-Description / Goal:	A registered user lists an item for sale, sets a price with an optional negotiation floor, and completes the sale through the platform's secure transaction flow.
Preconditions:	User is logged in. Item is assessed and listed (see UC-9). Item is marked as 'For Sale'.
Postconditions:	Item is sold and marked as unavailable. Funds are transferred to escrow pending delivery confirmation. Dynamic Sync removes item from all active listings.
Error Situations – System State in the Event of an Error:	Item is simultaneously engaged in a swap – System locks the item and rejects the sale attempt until the swap is resolved.
Actor(s) / Initiator(s):	Registered User (Seller), Registered User (Buyer)
Trigger:	A buyer initiates a purchase or a seller receives an offer on their listed item.
Standard Process (Main Success Scenario):	(1) Seller lists the item with a set price and configures a negotiation floor (minimum acceptable price). (2) Buyer views the listing and clicks 'Buy Now' or submits a price offer. (3) [If offer] System checks if the offer meets the Seller's auto-accept threshold. (4) Offer is accepted (manually or automatically) – Sale proceeds. (5) System triggers Agreement Lock-in: item description and price are frozen. (6) Buyer completes payment – Funds are held in Escrow. (7) Seller ships the item (see UC-19). (8) Buyer confirms receipt – Funds are released to Seller. (9) System updates Seller's trust score and Eco-Credits.
Alternative Processes (Alternative/Other Scenarios):	(2') Multiple items from same seller – System applies bundle discount automatically. (3') Offer is below auto-decline threshold – System automatically rejects the offer.

Describing Use Cases [[Buy Item](#)]

ID Number – Name:	UC-15 – Buy Item
Short-Description / Goal:	A registered user purchases an item from the marketplace, with an optional price negotiation phase, bundle discount for multiple items from the same seller, and secure payment via Escrow.
Preconditions:	User is logged in. Item is listed as 'For Sale' and not locked in another transaction. Buyer has a valid payment method.
Postconditions:	Purchase is confirmed. Item is locked and removed from the marketplace. Funds are held in Escrow until delivery is confirmed.
Error Situations – System State in the Event of an Error:	Payment fails – System notifies the buyer, order is cancelled, and item is re-listed.
Actor(s) / Initiator(s):	Registered User (Buyer), Registered User (Seller), Payment Gateway
Trigger:	Buyer clicks 'Buy Now' or 'Make an Offer' on an item listing.
Standard Process (Main Success Scenario):	(1) Buyer clicks 'Buy Now' on a listing. (2) System displays the order summary (item price + shipping cost + platform fee). (3) Buyer selects a payment method and enters payment details. (4) System validates payment details. (5) Payment is confirmed – Funds are transferred to Escrow. (6) System notifies the Seller to prepare and ship the item. (7) Seller ships the item. (8) Buyer confirms receipt. (9) Escrow releases funds to Seller.
Alternative Processes (Alternative/Other Scenarios):	(1') Buyer clicks 'Make an Offer' – Negotiation flow is triggered (see UC-13 Negotiate Price). (2') Buyer has multiple items from same seller – System auto-applies bundle discount to the order total. (4') Payment details invalid – System highlights errors and prompts correction.

Describing Use Cases [[Swap Item](#)]

ID Number – Name:	UC-16 – Swap Item
Short-Description / Goal:	Two registered users propose and agree to exchange items. A value balancer handles price differences, a 48-hour auto-cancel timer prevents stale offers, and an escrow secures any cash top-up.
Preconditions:	Both users are logged in. Both items are listed as 'For Swap' and not locked in another transaction.
Postconditions:	Swap is agreed and locked in. Both items are removed from the marketplace. Shipping is arranged for both parties. [If cash top-up] Difference is held in Escrow and transferred upon delivery confirmation.
Error Situations – System State in the Event of an Error:	No response within 48 hours – System auto-cancels the offer and unlocks both items.
Actor(s) / Initiator(s):	Registered User (Swapper A), Registered User (Swapper B), Moderator, Shipping
Trigger:	Swapper A clicks 'Propose Swap' on Swapper B's listing.
Standard Process (Main Success Scenario):	(1) Swapper A selects their item(s) to offer in exchange for Swapper B's item. (2) System calculates the value difference between items. (3) If value gap exists – System notifies both parties and suggests a cash top-up or an additional item. (4) Swap offer is sent to Swapper B. (5) Swapper B accepts – Moderator reviews and approves the swap. (6) Agreement Lock-in is triggered: descriptions and terms are frozen. (7) Dynamic Sync removes both items from the marketplace. (8) Both parties arrange shipping. (9) Both confirm receipt – Swap is marked as complete; Eco-Credits and trust scores are updated.
Alternative Processes (Alternative/Other Scenarios):	(3') Swapper A adds an extra item to bridge the value gap – Multi-item offer flow is triggered. (5') Swapper B rejects – System notifies Swapper A and unlocks Swapper A's items.

Describing Use Cases [[Calculate Carbon Savings](#)]

ID Number – Name:	UC-17 – Calculate Carbon Savings
Short-Description / Goal:	The system automatically calculates the estimated CO ₂ savings for each completed transaction based on item material type and weight, displaying the result to users and contributing to platform-wide sustainability reports.
Preconditions:	A transaction (sale, swap, or upcycle) has been completed. Item material type and weight data are available in the listing.
Postconditions:	Carbon savings is calculated and attached to the transaction record. User's cumulative sustainability impact is updated on their profile. Data is available for platform-wide reports by the Data Analyst.
Error Situations – System State in the Event of an Error:	Material or weight data is missing – System uses category-level averages and flags the estimate as approximate.
Actor(s) / Initiator(s):	System (automated), Data Analyst
Trigger:	A transaction is marked as completed.
Standard Process (Main Success Scenario):	(1) Transaction is marked as complete. (2) System retrieves item material type and weight from the listing. (3) System applies the carbon and water savings algorithm (based on material type and avoided new production). (4) Calculated savings are stored in the transaction record. (5) Savings are displayed on the user's profile impact dashboard. (6) Eco-Credits are calculated and awarded based on the impact score. (7) Data is aggregated for the Data Analyst's market and sustainability reports.
Alternative Processes (Alternative/Other Scenarios):	(2') Material data is missing – System uses average values for the item's category and labels the result as an estimate.

Describing Use Cases [Secure Escrow]

ID Number – Name:	UC-18 – Secure Escrow
Short-Description / Goal:	The system holds a buyer's payment in a secure virtual vault after purchase confirmation and releases it to the seller only upon verified delivery. In case of non-delivery or dispute, funds are returned to the buyer.
Preconditions:	Buyer has confirmed and paid for an order. Payment Gateway has processed the transaction successfully.
Postconditions:	[Delivered & confirmed] Funds are released to Seller. [Dispute] Admin determines responsibility; funds are refunded or partially released accordingly.
Error Situations – System State in the Event of an Error:	Escrow system fails to hold funds – Transaction is cancelled and buyer is fully refunded immediately.
Actor(s) / Initiator(s):	Payment Gateway, System Admin, Registered User (Buyer), Registered User (Seller)
Trigger:	Buyer's payment is successfully processed by the Payment Gateway.
Standard Process (Main Success Scenario):	(1) Buyer completes payment. (2) Payment Gateway transfers funds to the Escrow vault. (3) System notifies Seller that payment is secured and order should be shipped. (4) Seller ships the item. (5) Buyer confirms receipt of the item. (6) System releases Escrow funds to the Seller. (7) Transaction is marked as complete.
Alternative Processes (Alternative/Other Scenarios):	(5') Buyer does not confirm receipt within the defined window – System escalates to Admin. (5'') Buyer reports non-delivery – Admin investigates; if confirmed, funds are refunded to Buyer and Seller's trust score is reduced. (5''') Item is returned – Rollback transaction is triggered and Escrow is reversed.

Describing Use Cases [[Ship an Order](#)]

ID Number – Name:	UC-19 – Ship an Order
Short-Description / Goal:	The Shipping actor processes an order by generating a shipping label with full item and recipient details, optimizing the delivery route to minimize carbon emissions, and updating the order status throughout transit until delivery is confirmed.
Preconditions:	Order is confirmed and payment is secured in Escrow. Seller has prepared the package. Buyer's delivery address is provided.
Postconditions:	Item is delivered to the buyer. Order status is updated to 'Delivered'. Escrow release is triggered.
Error Situations – System State in the Event of an Error:	Delivery route cannot be optimised – System uses the default route and logs the failure for review.
Actor(s) / Initiator(s):	Shipping, Seller, Buyer
Trigger:	Seller marks the order as 'Ready to Ship'.
Standard Process (Main Success Scenario):	(1) Seller marks the order as 'Ready to Ship'. (2) System generates a shipping label containing: buyer name, delivery address, item brief description, item ID, and order value. (3) System runs route optimization to select the most carbon-efficient delivery path. (4) Shipping actor picks up the package. (5) System sets order status to 'In Transit'. (6) Buyer can track the shipment in real time. (7) Package is delivered – Buyer confirms receipt. (8) System updates status to 'Delivered' and triggers Escrow release.
Alternative Processes (Alternative/Other Scenarios):	(4') [Swap Meet] Buyer and Seller are located nearby – System suggests a local meetup instead of shipping to eliminate emissions. (7') Buyer is unavailable for delivery – Shipping actor leaves a notice; status set to 'Delivery Attempted'.

Describing Use Cases [Report Delivery Failure]

ID Number – Name:	UC-20 – Report Delivery Failure
Short-Description / Goal:	A buyer reports that an order was not received or was received in an unacceptable condition. Admin investigates and determines responsibility, applying financial consequences to the responsible party.
Preconditions:	Order status is 'In Transit' or 'Delivered'. Buyer has not confirmed receipt within the expected window, or has flagged an issue.
Postconditions:	[Buyer at fault] Shipping cost is deducted from buyer; item is returned. [Seller at fault] Seller pays shipping cost; buyer is fully refunded; Seller's trust score is reduced.
Error Situations – System State in the Event of an Error:	Admin cannot determine responsibility – Funds are held in Escrow pending further evidence submission.
Actor(s) / Initiator(s):	Registered User (Buyer), System Admin, Shipping, Payment Gateway
Trigger:	Buyer clicks 'Report Delivery Issue' or delivery window expires without buyer confirmation.
Standard Process (Main Success Scenario):	(1) Buyer clicks 'Report Delivery Issue' and describes the problem. (2) System logs the report and notifies Admin. (3) Admin reviews shipping evidence (tracking history, seller's proof of dispatch). (4) Admin determines responsibility. (5a) [Seller responsible] System deducts shipping cost from Seller, refunds Buyer, and reduces Seller's trust score. (5b) [Buyer responsible] System deducts shipping cost from Buyer and arranges item return. (6) Escrow is settled according to Admin's decision.
Alternative Processes (Alternative/Other Scenarios):	(1') Delivery window expires with no buyer action – System auto-escalates to Admin after a defined period. (4') Both parties share responsibility – Admin splits costs proportionally.

Describing Use Cases [[Agreement Lock-in](#)]

ID Number – Name:	UC-21 – Agreement Lock-in
Short-Description / Goal:	Once both parties (buyer/seller or two swappers) reach a final agreement, the system automatically freezes the item description, price, and photos to prevent any post-agreement modifications.
Preconditions:	Both parties have agreed to a transaction (sale or swap). Transaction is in 'Pending Confirmation' state.
Postconditions:	Item description, price, and photos are locked and cannot be edited. Both parties are notified that the agreement is locked. Transaction proceeds to payment/shipping phase.
Error Situations – System State in the Event of an Error:	Lock-in system fails – Transaction is paused and both parties are notified to re-confirm before proceeding.
Actor(s) / Initiator(s):	System (automated), Registered User (Buyer/Seller/Swapper)
Trigger:	Both parties digitally confirm the transaction agreement.
Standard Process (Main Success Scenario):	(1) Both parties confirm agreement (Buyer clicks 'Confirm Purchase' or both Swappers accept the swap). (2) System sets item status to 'Agreement Locked'. (3) System freezes all editable fields: description, photos, price, and swap terms. (4) Any subsequent edit attempts by either party are blocked. (5) System notifies both parties that the agreement is now locked. (6) Transaction proceeds to the payment or shipping phase.
Alternative Processes (Alternative/Other Scenarios):	(4') A party attempts to edit a locked field – System displays a 'Field Locked – Agreement in Progress' message.

Describing Use Cases [[Dynamic Sync](#)]

ID Number – Name:	UC-22 – Dynamic Sync
Short-Description / Goal:	When an item enters an active agreement (sale or swap confirmed by both parties), the system automatically locks the item and removes it from the marketplace instantly, preventing it from appearing in any other listings or search results.
Preconditions:	Item is listed on the marketplace for sale or swap. Both parties have reached a confirmed agreement.
Postconditions:	Item is locked and no longer visible on the marketplace. Any other pending offers on the item are automatically cancelled and their initiators notified.
Error Situations – System State in the Event of an Error:	Sync fails and item remains visible in another listing – System raises an alert and re-triggers the sync; auto-cancels all pending offers on the item
Actor(s) / Initiator(s):	System (automated)
Trigger:	Both parties confirm a transaction agreement
Standard Process (Main Success Scenario):	(1) Both parties confirm the agreement. (2) System instantly locks the item. (3) Item is removed from the marketplace and no longer appears in search or browse results. (4) All other pending offers for the item are automatically cancelled. (5) Affected users (those who had pending offers) are notified of the cancellation. (6) Item status is updated to 'In Transaction' and hidden from the marketplace.
Alternative Processes (Alternative/Other Scenarios):	(4') No other pending offers exist — System skips cancellation step and proceeds directly.

Describing Use Cases [[Swap Meet Finder](#)]

ID Number – Name:	UC-22 – Meet Finder
Short-Description / Goal:	The system identifies nearby users based on both parties' locations and, when the distance is within a defined threshold, suggests an in-person meetup for a swap instead of shipping — eliminating shipping emissions. The marketplace also prioritizes showing listings from users who are geographically close to the browsing user.
Preconditions:	Both users have granted location access. A swap offer is in progress, OR a user is browsing the marketplace.
Postconditions:	[Swap] Both parties arrange an in-person exchange; no shipping is generated. [Browse] User sees nearby listings ranked higher in results. Carbon savings from avoided shipping are calculated and credited.
Error Situations – System State in the Event of an Error:	Location data is unavailable for one or both users — System defaults to standard shipping flow and notifies users that Meet Finder is unavailable.
Actor(s) / Initiator(s):	System (automated), Registered User
Trigger:	A swap offer is initiated between two users, OR A user opens the marketplace (for proximity-based ranking).
Standard Process (Main Success Scenario):	(1) Swap offer is initiated between Swapper A and Swapper B. (2) System calculates the distance between both users' locations. (3) Distance is within the defined threshold — System suggests a local meetup instead of shipping. (4) Both parties agree to meet in person. (5) System marks the swap as "Local Meetup" — no shipping label is generated. (6) System marks the swap as complete and calculates carbon savings from avoided shipping.
Alternative Processes (Alternative/Other Scenarios):	(3') Distance exceeds the threshold — System proceeds with the standard shipping flow. (4') One party declines the meetup — System falls back to standard shipping.